

TEBOGO PHINITHI

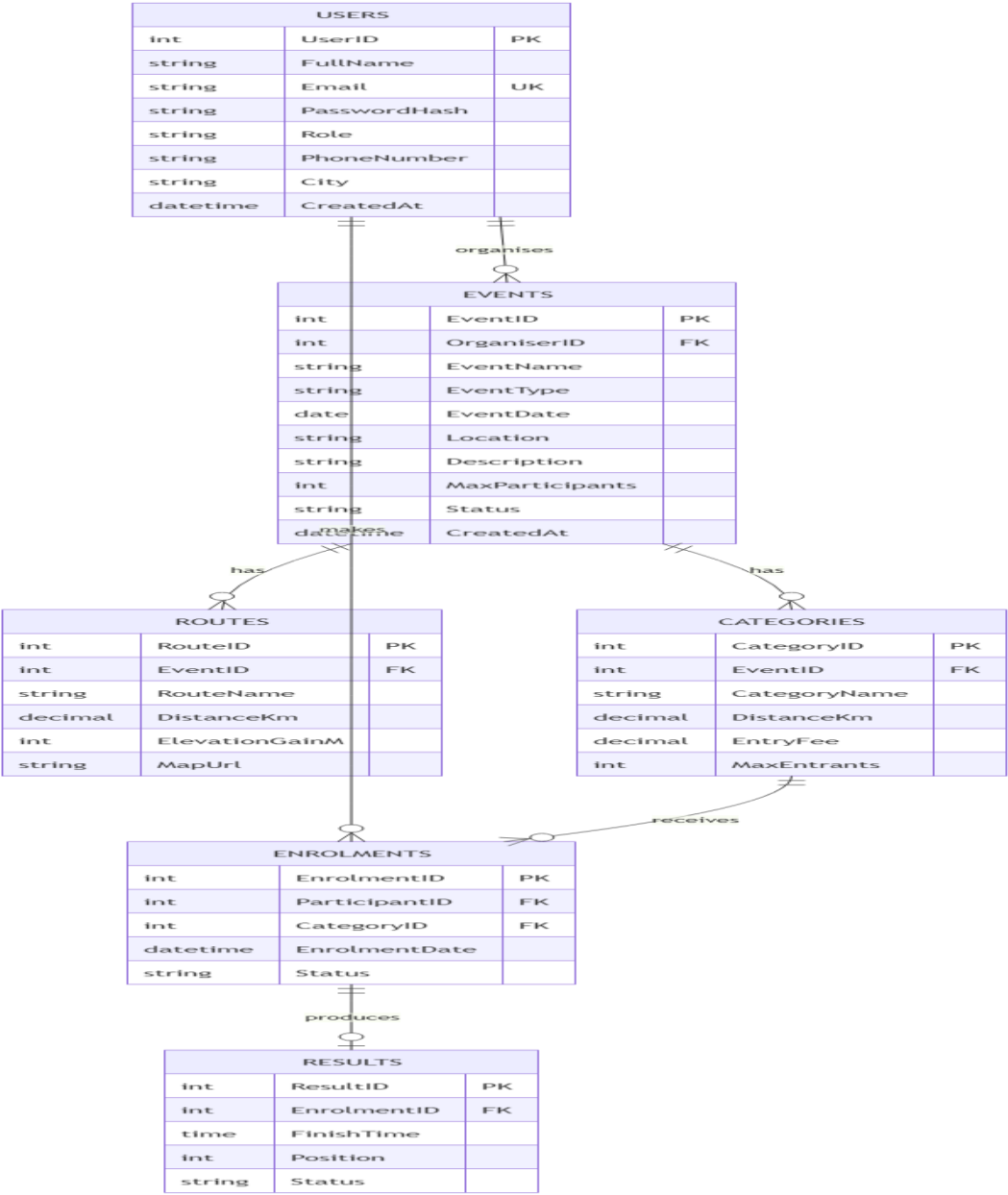
PROGRAMMING ASSINGNMENT PART 1

GROUP 4

ST10486400

04 SEPTEMBER 2026

SECTION A



SECTION B

RACEDAY- API-ENDPOINT PLAN

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as Organiser or Participant	None (public)	{ fullName, email, password, role, phoneNumber, city }	201 Created – new user (no password) 400 Bad Request – invalid input 409 Conflict – email already registered
POST	/api/auth/login	Authenticates a user and returns a token	None (public)	{ email, password }	200 OK – token + user profile 401 Unauthorized – invalid credentials
GET	/api/users/me	Returns the logged-in user's profile	Any (logged in)	None	200 OK – user profile 401 Unauthorized
PUT	/api/users/me	Updates the logged-in user's profile	Any (logged in)	{ fullName, phoneNumber, city }	200 OK – updated profile 400 Bad Request 401 Unauthorized
GET	/api/events	Lists all upcoming events (searchable/filterable)	None (public)	None	200 OK – array of events
GET	/api/events/{id}	Returns a single event with its categories	None (public)	None	200 OK – event details 404 Not Found
POST	/api/events	Creates a new event	Organiser	{ eventName, eventType, eventDate, location, description, maxParticipants }	201 Created – new event 400 Bad Request 401 Unauthorized 403 Forbidden
PUT	/api/events/{id}	Updates an event owned by the logged-in organiser	Organiser	{ eventName, eventType, eventDate, location, description, maxParticipants, status }	200 OK – updated event 400 Bad Request 403 Forbidden 404 Not Found
DELETE	/api/events/{id}	Deletes an event owned by the logged-in organiser	Organiser	None	204 No Content 403 Forbidden 404 Not Found
GET	/api/events/{id}/categories	Lists categories for an event	None (public)	None	200 OK – array of categories 404 Not Found
POST	/api/events/{id}/categories	Adds a category to an event	Organiser	{ categoryName, distanceKm, entryFee, maxEntrants }	201 Created – new category 400 Bad Request 403 Forbidden 404 Not Found
PUT	/api/categories/{id}	Updates a category	Organiser	{ categoryName, distanceKm, entryFee, maxEntrants }	200 OK – updated category 400 Bad Request 403 Forbidden 404 Not Found
DELETE	/api/categories/{id}	Deletes a category	Organiser	None	204 No Content 403 Forbidden 404 Not Found
GET	/api/events/{id}/routes	Lists route information for an event	None (public)	None	200 OK – array of routes 404 Not Found
POST	/api/enrolments	Enrols the logged-in participant in a category	Participant	{ categoryId }	201 Created – new enrolment 400 Bad Request 401 Unauthorized 404 Not Found (category) 409 Conflict (already enrolled / category full)
GET	/api/enrolments/me	Lists the logged-in participant's enrolments (performance history)	Participant	None	200 OK – array of enrolments with results 401 Unauthorized
DELETE	/api/enrolments/{id}	Cancels the logged-in participant's own enrolment	Participant	None	204 No Content 403 Forbidden 404 Not Found
GET	/api/events/{id}/enrolments	Lists all enrolments for an event (for the organiser to manage)	Organiser	None	200 OK – array of enrolments 403 Forbidden 404 Not Found
POST	/api/results	Captures a result for a participant's enrolment	Organiser	{ enrolmentId, finishTime, position, status }	201 Created – new result 400 Bad Request 403 Forbidden 404 Not Found 409 Conflict (result already exists)
PUT	/api/results/{id}	Updates a captured result	Organiser	{ finishTime, position, status }	200 OK – updated result 400 Bad Request 403 Forbidden 404 Not Found
GET	/api/results/{id}	Returns a single result	Any (logged in)	None	200 OK – result 404 Not Found

SECTION C

IF DB_ID('RaceDayDB') IS NOT NULL

BEGIN

ALTER DATABASE RaceDayDB SET SINGLE_USER WITH ROLLBACK IMMEDIATE;

DROP DATABASE RaceDayDB;

END

CREATE DATABASE RaceDayDB;

USE RaceDayDB;

CREATE TABLE Users (

UserID INT IDENTITY(1,1) PRIMARY KEY,

FullName VARCHAR(100) NOT NULL,

Email VARCHAR(150) NOT NULL UNIQUE,

PasswordHash VARCHAR(256) NOT NULL,

[Role] VARCHAR(20) NOT NULL,

PhoneNumber VARCHAR(20) NULL,

City VARCHAR(50) NULL,

CreatedAt DATETIME2 NOT NULL DEFAULT GETDATE(),

CONSTRAINT CK_Users_Role CHECK ([Role] IN ('Organiser', 'Participant'))

);

```

CREATE TABLE Events (
    EventID    INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserID INT      NOT NULL,
    EventName   VARCHAR(150) NOT NULL,
    EventType   VARCHAR(20) NOT NULL,
    EventDate   DATE      NOT NULL,
    [Location]  VARCHAR(100) NOT NULL,
    [Description] VARCHAR(1000) NULL,
    MaxParticipants INT      NULL,
    [Status]    NVARCHAR(20) NOT NULL DEFAULT 'Upcoming',
    CreatedAt   DATETIME2   NOT NULL DEFAULT GETDATE(),
    CONSTRAINT FK_Events_Organiser FOREIGN KEY (OrganiserID) REFERENCES
Users(UserID),
    CONSTRAINT CK_Events_Type CHECK (EventType IN ('Running', 'Walking', 'Cycling')),
    CONSTRAINT CK_Events_Status CHECK ([Status] IN ('Upcoming', 'Completed',
'Cancelled'))
);

```

```

CREATE TABLE Categories (
    CategoryID INT IDENTITY(1,1) PRIMARY KEY,
    EventID    INT      NOT NULL,

```

```
CategoryName VARCHAR(100) NOT NULL,  
DistanceKm DECIMAL(5,2) NOT NULL,  
EntryFee DECIMAL(8,2) NOT NULL DEFAULT 0,  
MaxEntrants INT NULL,  
CONSTRAINT FK_Categories_Event FOREIGN KEY (EventID) REFERENCES  
Events(EventID),  
CONSTRAINT UQ_Categories_EventName UNIQUE (EventID, CategoryName)  
);
```

```
CREATE TABLE Routes (  
RouteID INT IDENTITY(1,1) PRIMARY KEY,  
EventID INT NOT NULL,  
RouteName VARCHAR(100) NOT NULL,  
DistanceKm DECIMAL(5,2) NOT NULL,  
ElevationGainM INT NULL,  
MapUrl VARCHAR(300) NULL,  
CONSTRAINT FK_Routes_Event FOREIGN KEY (EventID) REFERENCES Events(EventID)  
);
```

```
CREATE TABLE Enrolments (  
EnrolmentID INT IDENTITY(1,1) PRIMARY KEY,  
ParticipantID INT NOT NULL,
```

```

CategoryID INT NOT NULL,
EnrolmentDate DATETIME2 NOT NULL DEFAULT GETDATE(),
[Status] VARCHAR(20) NOT NULL DEFAULT 'Confirmed',
CONSTRAINT FK_Enrolments_Participant FOREIGN KEY (ParticipantID) REFERENCES
Users(UserID),
CONSTRAINT FK_Enrolments_Category FOREIGN KEY (CategoryID) REFERENCES
Categories(CategoryID),
CONSTRAINT UQ_Enrolments_ParticipantCategory UNIQUE (ParticipantID, CategoryID),
CONSTRAINT CK_Enrolments_Status CHECK ([Status] IN ('Confirmed', 'Cancelled'))
);

```

```

CREATE TABLE Results (
ResultID INT IDENTITY(1,1) PRIMARY KEY,
EnrolmentID INT NOT NULL UNIQUE,
FinishTime TIME NULL,
[Position] INT NULL,
[Status] VARCHAR(20) NOT NULL DEFAULT 'Finished',
CONSTRAINT FK_Results_Enrolment FOREIGN KEY (EnrolmentID) REFERENCES
Enrolments(EnrolmentID),
CONSTRAINT CK_Results_Status CHECK ([Status] IN ('Finished', 'DNF', 'DQ'))
);

```

```
INSERT INTO Users (FullName, Email, PasswordHash, [Role], PhoneNumber, City)
VALUES
('Thabo Nkosi', 'thabo.nkosi@raceday-events.co.za', 'HASHED_PWD_1', 'Organiser',
'0821234567', 'Johannesburg'),
('Sarah van der Merwe', 'sarah.vdm@raceday-events.co.za', 'HASHED_PWD_2',
'Organiser', '0834567890', 'Cape Town');
```

```
INSERT INTO Users (FullName, Email, PasswordHash, [Role], PhoneNumber, City)
VALUES
('Lindiwe Dlamini', 'lindiwe.dlamini@example.co.za', 'HASHED_PWD_3', 'Participant',
'0712345678', 'Soweto'),
('Johan Botha', 'johan.botha@example.co.za', 'HASHED_PWD_4', 'Participant',
'0723456789', 'Pretoria');
```

```
INSERT INTO Events (OrganiserID, EventName, EventType, EventDate, [Location],
[Description], MaxParticipants, [Status])
VALUES
(1, 'Soweto Marathon', 'Running', '2026-11-01', 'Soweto', 'Annual road running event
through the streets of Soweto.', 5000, 'Upcoming'),
(2, 'Cape Town Cycle Tour', 'Cycling', '2026-10-15', 'Cape Town', 'Scenic cycling tour
around the Cape Peninsula.', 8000, 'Upcoming'),
(1, 'Pretoria Fun Walk', 'Walking', '2026-09-20', 'Pretoria', 'Community fun walk in support
of local charities.', 2000, 'Upcoming');
```

```
INSERT INTO Categories (EventID, CategoryName, DistanceKm, EntryFee, MaxEntrants)
VALUES
(1, '10km Individual', 10.00, 150.00, 2000),
(1, '42km Full Marathon', 42.20, 350.00, 1500),
(2, '109km Cycle Tour', 109.00, 550.00, 6000),
(2, '35km Mini Tour', 35.00, 300.00, 2000),
(3, '5km Fun Walk', 5.00, 50.00, 2000);
```

```
INSERT INTO Routes (EventID, RouteName, DistanceKm, ElevationGainM, MapUrl)
VALUES
(1, 'Soweto Marathon Main Route', 42.20, 320, 'https://maps.example.co.za/soweto-marathon'),
(2, 'Cape Peninsula Loop', 109.00, 980, 'https://maps.example.co.za/cape-cycle-tour'),
(3, 'Pretoria CBD Loop', 5.00, 40, 'https://maps.example.co.za/pretoria-fun-walk');
```

```
INSERT INTO Enrolments (ParticipantID, CategoryID, [Status])
VALUES
(3, 2, 'Confirmed'),
(4, 3, 'Confirmed'),
(3, 5, 'Confirmed');
```

```
INSERT INTO Results (EnrolmentID, FinishTime, [Position], [Status])  
VALUES  
(1, '03:45:12', 152, 'Finished');
```

```
SELECT e.EventName, e.EventType, e.EventDate, e.[Location], u.FullName AS Organiser  
FROM Events e  
JOIN Users u ON e.OrganiserID = u.UserID;
```

```
SELECT e.EventName, c.CategoryName, c.DistanceKm, c.EntryFee  
FROM Categories c  
JOIN Events e ON c.EventID = e.EventID  
ORDER BY e.EventName;
```

```
SELECT u.FullName AS Participant, e.EventName, c.CategoryName,  
       en.[Status] AS EnrolmentStatus, r.FinishTime, r.[Position]  
FROM Enrolments en  
JOIN Users u ON en.ParticipantID = u.UserID
```

```
JOIN Categories c ON en.CategoryID = c.CategoryID
JOIN Events e ON c.EventID = e.EventID
LEFT JOIN Results r ON r.EnrolmentID = en.EnrolmentID
ORDER BY u.FullName;
```

```
SELECT e.EventName, ro.RouteName, ro.DistanceKm AS RouteDistance,
ro.ElevationGainM
FROM Events e
JOIN Routes ro ON ro.EventID = e.EventID
WHERE e.EventName = 'Soweto Marathon';
```

```
SELECT e.EventName, COUNT(en.EnrolmentID) AS TotalEnrolments
FROM Events e
JOIN Categories c ON c.EventID = e.EventID
LEFT JOIN Enrolments en ON en.CategoryID = c.CategoryID
GROUP BY e.EventName;
```